

Research article

Blockchain-based federated learning with homomorphic encryption for privacy-preserving healthcare data sharing



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ABSTRACT

Healthcare data is often fragmented across various institutions due to its highly sensitive and private nature. In this sense, hospitals and clinics maintain electronic health records (EHRs) independently; hence, valuable data is siloed within individual organizations, preventing comprehensive analysis that could benefit from diverse data sources. Federated learning (FL) addresses these challenges by enabling the training of a shared global model using data distributed across multiple institutions without moving the data from its source. By leveraging FL, healthcare institutions can combine their data assets to improve predictive analytics, personalized medicine, and overall healthcare outcomes, ultimately benefiting patients and the healthcare system. However, the current FL model with a central server presents several challenges within healthcare, including the risk of malicious attacks, regulatory compliance, and privacy vulnerabilities. To overcome these issues, this paper introduces the FL framework with blockchain and homomorphic encryption (HE). Our framework aims to minimize the role of the central server, enable collaborative model training across healthcare organizations, and enhance data security and privacy. In this sense, blockchain ensures the integrity and transparency of the process, while homomorphic encryption ensures that the data remains private. This framework can potentially enable institutions to enrich medical knowledge while securely keeping patient data collaboratively and facilitating healthcare analytics in practical settings.

1. Introduction

In recent years, the availability of healthcare data from various sources, like clinical institutions, pharmaceutical industries, and insurance companies, has surged [1]. It offers opportunities to develop data-driven analytics to improve the quality of the healthcare system. However, healthcare data is often fragmented due to its highly sensitive and private nature. Different hospitals and institutions typically only have access to their own patient records through electronic health records (EHRs). These records contain sensitive protected health information (PHI), regulated by stringent laws such as the Health Insurance Portability and Accountability Act (HIPAA) [2]. This fragmentation and regulation pose significant challenges for advanced healthcare data analytics, which require enormous amounts of training data to be used. As a result, valuable data is isolated within individual organizations, limiting the ability to conduct thorough analysis that could leverage data from multiple sources.

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To address these challenges, federated learning (FL) has gained popularity in healthcare for its ability to train shared global models while keeping data within local institutions, thus enhancing data security and privacy [3]. Unlike traditional machine learning (ML) methods, FL transmits trained models instead of raw data, enabling collaborative model training without exposing sensitive information. In this sense, healthcare institutions can combine their data assets to improve overall healthcare outcomes, ultimately benefiting patients and the healthcare system. Moreover, the FL approach aligns with stringent data privacy regulations like HIPAA, making it ideal for advancing predictive diagnostics, personalized medicine, medical research, and healthcare analytics. Thus, FL facilitates innovation in improving care delivery, as well as overcomes data fragmentation and regulatory challenges.

Despite its benefits, the existing FL model with a centralized server poses several challenges within healthcare [4]. These challenges include the risk of malicious attacks, regulatory compliance, and privacy vulnerabilities. Moreover, FL's reliance on a central server for model aggregation introduces a single point of failure, raising concerns about data security and privacy. Hostile users may exploit this vulnerability to conduct various attacks, such as membership inference and poisoning attacks, which can compromise the development of the global model and undermine overall system security and effectiveness [5]. Consequently, healthcare entities may hesitate to participate in collaborative FL training due to these concerns. Therefore, addressing these challenges is essential to fully harnessing FL's potential in healthcare and ensuring the security and privacy of sensitive patient data.

Several privacy-preserving techniques have been proposed to address these challenges. Secure aggregation ensures that the server only learns aggregated updates rather than individual client updates [6]. However, it still allows access to aggregated gradients, which can leak information about the underlying model. Differential privacy (DP) is another commonly used technique in FL that enhances security by adding noise to model updates, thereby protecting individual data [7]. While DP can be easily integrated and implemented, it suffers from weaker privacy guarantees, which can lead to accuracy degradation. Additionally, DP exposes raw gradients to the central server, potentially leading to privacy breaches [8]. Secure multi-party computation (MPC) enables multiple clients to collaboratively compute a function over their private data without revealing it. However, MPC requires complex synchronization protocols and significant computational overhead, making it inefficient for geo-distributed FL [9].

Motivated by the challenges and concerns mentioned above, this paper introduces the FL framework enhanced with blockchain and homomorphic encryption (HE). Our framework aims to minimize the role of the central server, enable collaborative model training across healthcare organizations, and strengthen data security and privacy. HE facilitates encrypted gradient aggregation, ensuring that the central server never has access to raw model updates. Unlike existing approaches (such as secure aggregation [6], DP [7], and MPC [9]), HE offers strong privacy guarantees without compromising learning accuracy and can be integrated into existing FL frameworks. In this context, HE protects data privacy by allowing multiple participants, such as healthcare entities, to engage in collaborative FL model training while keeping their data encrypted. Additionally, blockchain provides decentralized trust, transparency, and immutability, removing the need for a trusted central authority and preventing data manipulation. Blockchain technology also ensures process integrity and transparency by recording all transactions on a distributed ledger. Furthermore, it can establish a decentralized incentive system to promote participation in collaborative training among healthcare institutions. In short, the combination of HE and blockchain ensures secure, verifiable, and tamper-resistant FL training, making our framework more robust than current methods. Ultimately, this framework has the potential to transform healthcare analytics by enabling collaborative institutions to improve medical knowledge while securely handling patient data.

The structure of this paper unfolds as follows: Section 2 offers a brief background on homomorphic encryption, blockchain technology, and federated learning, laying the foundation for understanding our proposed approach. Section 3 reviews related works. In Section 4, we present our concept designed to address security and privacy challenges in healthcare systems. Section 5 analyzes the numerical results obtained from simulations conducted to evaluate the efficacy of our framework. Lastly, Section 6 concludes the paper.

2. Preliminaries

2.1. Traditional healthcare data sharing

As technology progresses, there has been a surge in the amount of digital data as it has become easier to digitize various kinds of information, including the medical-related data in healthcare field. This has brought forward a new opportunity to process and analyze the data through myriad data analytics methods, including various machine learning techniques, which can be a catalyst in generating insights and potentially enhance the quality of the overall healthcare outcome—such as through improved predictive analytics, personalized medicine, more effective management, and even contributing to various findings, e.g., of more potent medication, more proper treatments, or even breakthrough discoveries.

However, due to the highly sensitive and private nature of healthcare data such as patients' records, care providers (e.g., hospitals and clinics) manage these data privately within their organization, i.e., in a *silo* approach. This is to preserve the anonymity of individual patients, which should be maintained in the highest possible standard [10]. On the other hand, as data availability through sharing has now been regarded as essential in enhancing the clinical treatments and medical research [11], there has also been a paradigm shift in the medical-related research towards collaboration between multiple institutions, but required to be conducted in a very strict manner to comply with the data privacy and protection laws. This has its own shortcoming since different regions and institutions may have its own regulations, potentially results in data law conflict that are difficult to resolve, which even more hinder the full potential of data utility across different institutions.

Legacy approach resorts to bespoke data sharing contracts which often requires lengthy administration process, further reducing efficiency and deterring the collaboration initiatives [12]. This is unfortunate considering the potential of collaboration to obtain much more diverse data may promote expedition of data analysis and more accurate results, which in turn, benefit patients, and the overall healthcare system, and health-related science in general. In contrast, global collaboration and extensive data sharing may be highly beneficial in advancing the medical fields, such as research in critical illness or rare diseases with unknown cures, e.g., previously COVID-19, HIV/AIDS, early detection of cancer, brain tumors, to mention a few.

In addition, at the case where collaboration can be conducted via the bespoke contracts, traditional data sharing approaches often relies on conventional anonymization techniques such as by suppressing directly identifying attributes or by generalizing, aggregating, or randomizing quasi-identifying attributes in individual patient records [12], which is at risk due to the increasing threat of reidentification attacks from anonymized data [12–15].

2.2. Homomorphic encryption

Homomorphic encryption (HE) is one type of encryption scheme that enables arithmetic operations to be performed directly on encrypted data or ciphertext without the need for decryption in advance. Computed outputs are likewise encrypted, with identical (or almost identical) results upon decryption, meaning that this scheme enables data processing without revealing the actual data [16]. This brings substantial implications as it allows third-party services (e.g., cloud computing services, cloud applications for data sharing and collaboration, and privacy-preserving ML services [17]) to offer more services to users without requiring to reveal users' sensitive information that compromises their privacy—a capability not satisfactorily offered by conventional encryption techniques [18].

An encryption technique is considered as homomorphic through an operation “ $\otimes$ ” if it satisfies Eq. (1), where E denotes the encryption algorithm while M denotes the set of all possible messages [18].

$$E(m_1) \otimes E(m_2) = E(m_1 \otimes m_2), \quad \forall m_1, m_2 \in M \quad (1)$$

Meanwhile, the HE property is described by Eq. (2), with Enc and Dec denoting encryption and decryption, respectively, and $f()$ is a function applied on actual values (i.e., the cleartexts) a and b using encryption key pk [16].

$$f(a, b) = Dec(Enc(pk, a), Enc(pk, b)) \quad (2)$$

Generally, HE schemes in literature can be classified under three types of schemes based on the number of allowed operations on the encrypted data [17,18]:

1. Partially Homomorphic Encryption (PHE): supports a single type of operation (i.e., only additions or only multiplications) for an unlimited number of times, meaning that there is no bound on the number of usages.
2. Somewhat Homomorphic Encryption (SWHE): permits both operations, but only for a limited number of times.
3. Fully Homomorphic Encryption (FHE) enables an unlimited number of operations to be performed an unlimited number of times.

2.3. Blockchain: Distributed ledger technology

In a nutshell, blockchain technology is a decentralized and distributed ledger system that eliminates the need for centralized management. Instead, it relies on a network of nodes working together to securely and immutably record transactions [19]. This decentralized approach minimizes the need for intermediaries, enhances transparency, and uses cryptographic methods to improve data security [20]. Originally designed to support cryptocurrencies like Bitcoin [21], blockchain technology has evolved into a promising solution for various industries due to its prominent features. Beyond its initial use for Bitcoin, blockchain has gained significant attention in sectors such as supply chain management, finance and banking, healthcare, and government. In these areas, it offers a reliable environment, enhances data security, and has the potential to reform data management [22].

A blockchain, as illustrated in Fig. 1, consists of the structure of a chain linked by hashes. Using Bitcoin as an instance, each block comprises two main parts, i.e., a header and a body. The block header contains essential metadata about the block and is composed of several components, including the difficulty value, the Merkle root, a timestamp, the hash of the previous block (parent hash), the version number, and nonce. Meanwhile, the block body comprises the actual data being recorded on the blockchain and consists of the number of transactions included in the block (i.e., transaction count), as well as the individual transactions from users, which are stored in a Merkle tree structure [24]. Together, these elements form a robust and secure structure. The block header summarizes the block's contents and ensures each block is cryptographically linked to the previous block, creating a continuous chain. The block body holds the transaction data, which can be efficiently verified and traced back to its source through the hash pointer in the header, whereas the timestamp ensures the blocks' correct order. Additionally, to secure data access and transmission, blockchain uses a consensus algorithm maintained by all nodes to generate and update data. Consequently, this design ensures data integrity, security, and transparency.

2.4. Federated learning

The Google AI team introduced Federated Learning [25] as a new communication-efficient optimization algorithm to address the challenges of the classical ML approach, particularly data privacy and communication efficiency. In traditional ML, the server

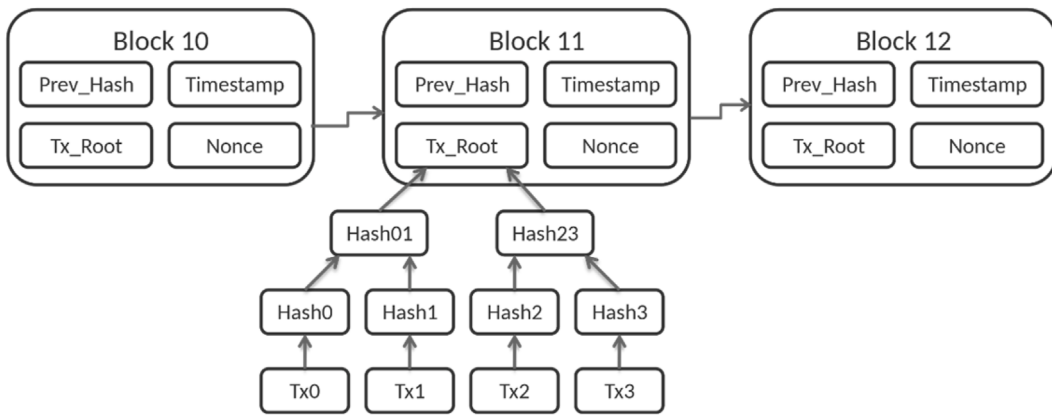


Fig. 1. Blockchain structure [23].

consistently handles model training, performing both data training and aggregation in a centralized manner, where data is collected and processed on a central server. In contrast, FL allows model training to occur directly on devices, such as smartphones or IoT devices, without requiring raw data to be uploaded to a central server. In this sense, users only send the essential information needed to update learning objectives while keeping sensitive data locally [26]. By keeping sensitive user data on local devices, FL significantly mitigates privacy concerns by preventing certain attacks that attempt to gain unauthorized access to local training data, making it particularly valuable in sensitive fields such as healthcare and finance [27]. Thus, due to its advantages, FL has been applied in various learning domains and applications, including next-word prediction [28], enhancing vocabulary and language models [29], developing predictive models for patient care [30], and facilitating low-latency vehicle-to-vehicle communication for autonomous driving [31].

3. Related works

In the context of the existing healthcare environment, multiple challenges frequently deter the exchange of information and seamless collaboration among healthcare institutions and stakeholders. This is understandable considering the inherently sensitive nature of healthcare data results in the paramount concerns of data privacy and security [22]. The need to comply with safeguarding-yet-complicated personal data regulatory frameworks, including the ones specific to healthcare (e.g., HIPAA of the United States [2], The Health Records Act (HRA) of Australia, and Health Information Privacy Code of New Zealand), further augment the intricacy of data-sharing initiatives. The concept of FL was introduced as one of the possible solutions to resolve the challenges while navigating through the requirement of healthcare data protection. FL offers a viable method for medical institutions to collaborate and share insights derived from Electronic Health Record (EHR) data, while maintaining data privacy by not directly sharing the data itself [32].

In recent years, there have been several works in the literature that proposed employing HE (and its variant) for a more secure FL, e.g., [16,17,33–35]. In [17], the authors proposed to utilize FHE to alleviate privacy and security concerns of FL over insecure communication channels for the use case of IoT-enabled smart city communications. In particular, the authors proposed a flexible design of FL and FHE combination which is adaptable to different IoT environments, with four distinct scenarios which are then examined as a proof-of-concept (PoC) of the work. These four scenarios are then evaluated against an IoT dataset benchmark in terms of their security and efficiency, which show improvements around 80.15 to 89.98% in terms of communication overhead compared to the baseline scenario. In [33], the authors proposed a multiparty privacy-preserving ML framework entitled PFMLP based on PHE and FL to perform ML tasks without revealing sensitive data. To enhance the efficiency of their technique over the standard HE which induces relatively high computational overhead (e.g., additional overhead in encryption and decryption during training), they proposed to employ the improved Paillier algorithm, yielding an increase in the training speed of 25%–28%.

Related to the healthcare sector, in [16], the authors proposed the use of HE and FL to perform a privacy-preserving ML task in medical data to safeguard against adversarial attack by establishing a secure multiparty computation protocol. In particular, the authors employ the method in the training of Convolutional Neural Network (CNN) for the use case of COVID-19 detection with real-world dataset and evaluate their method based on its performance, showing a comparative performance in terms of accuracy, F1 score, precision, and recall. Note that they observe the considerable tradeoff for the number of clients and the execution time. In [36], the authors introduced a homomorphic outsourcing computing toolkit for secure predictive computation, utilizing the Paillier encryption system and Lagrange interpolation to perform basic operations and encrypted comparisons. Similarly, [37] proposed comparable homomorphic encryption (CompHE) schemes that enable efficient ciphertext comparison among multiple users using Lagrange's interpolation technique, enhancing security through collusion resistance in the semi-honest model while requiring fewer modular exponentiations for comparison. These schemes also demonstrate applications in privacy-preserving COVID-19 contact tracing and social distance control. In [38], the authors proposed a secure data fitting scheme for medical IoT based on the CKKS

Table 1
Summary of notations.

Notations	Descriptions
H	Number of users (i.e., hospitals), $H = \{h_1, h_2, h_3, \dots, h_n\}$
C	Number of clusters, $C = \{C_1, C_2, C_3, \dots, C_n\}$
E	Number of edge servers, $E = \{E_1, E_2, E_3, \dots, E_n\}$
ψ^{in}	The initial parameters of the global model
key_E^{pk}, key_E^{sk}	Public-private key pairs of edge servers
key_h^{pk}, key_h^{sk}	Public-private key pairs of participating hospitals
D	Local datasets, $D_i = \{D_1, D_2, D_3, \dots, D_n\}$
t	Number of iteration
δ	The similarity of hospitals' data distribution
θ_i^*	The stationary solution of hospital i
ψ_h^i	The trained local model
$\nabla F_h(\psi)$	The average of model parameters update
$Enc_HE(\psi_h^i)$	The encrypted local model with homomorphic encryption
ψ_{gbl}^i	The new global model for the next iteration $t + 1$
$Enc(\psi_h^i)$	The encryption function of ψ_h^i
$Dec(\psi_{gbl}^i)$	The decryption function of ψ_{gbl}^i
R_h^i	Rewards of contribution for participating hospitals

homomorphic encryption algorithm, which encrypts the KAGGLE-HDP Heart Disease Prediction dataset and calculates the data's weight and deviation using gradient descent. The scheme achieves a recognition accuracy of 96.7%, offering high accuracy and improved privacy protection. Additionally, the authors of [35] presented an FL-PHE combination within a distributed data fabric architecture to establish a secure medical image analysis that complies with data protection and privacy laws, enabling multiple clients (e.g., hospitals and medical data centers) to collaborate in training a machine-learning model without exchanging raw data, thus not requiring centralized repository. The effectiveness of their method are tested on the use case of pituitary tumor classification, which yields an accuracy of 83.31%.

In terms of the integration of blockchain to the FL-HE itself, Wang et al. [39] proposed BPFL, a FL-HE scheme leveraging consortium blockchain as its underlying distributed framework for the case of the Internet of Vehicle (IoV). In particular, the authors improve the existing aggregation rules called Multi-Krum and combine it with HE to achieve ciphertext-level model aggregation and filtering to enable the verifiability of the local models while being privacy-preserved. Further, for the blockchain side, they also develop a reputation-based incentive mechanism to encourage users in IoV to actively and honestly participate in the FL scheme. On the other hand, the work from [40] proposed a FL approach combining blockchain, HE, and reputation mechanisms to ensure privacy and security in distributed data interactions. Edge nodes train ciphertext models using HE, with contributions evaluated through a reputation system and verified on the blockchain. This approach incentivizes participation, protects data privacy, and addresses security risks, while a joint aggregation and optimization algorithm enhances learning accuracy and performance.

4. Proposed scheme

This research aims to present a secure FL approach by utilizing HE and blockchain to tackle security and privacy concerns in the healthcare system. Our scheme employs a cross-silo FL setting, which is suitable for healthcare applications where participants, such as hospitals, are consistently available to contribute to collaborative training in each round of computation. Blockchain ensures process integrity and transparency by recording all transactions in its distributed ledger and provides a decentralized incentive mechanism to encourage participation in collaborative training among healthcare institutions [41]. Meanwhile, HE safeguards data privacy by allowing multiple participants, such as healthcare entities, to engage in collaborative FL model training while keeping their data encrypted. Table 1 shows the summary of notations used in this paper.

4.1. System entities and components

Before presenting the design architecture, we provide definitions of the relevant terms used in this paper.

- (i) **Users:** The users consist of participating medical institutions (i.e., hospitals in our case) that possess large amounts of raw medical data. In this setup, hospitals act as data providers, where they perform local model training using their own data. Additionally, encryption is conducted locally by the users before the data is uploaded to the edge server.
- (ii) **Local datasets:** The data is stored and processed locally at each participating hospital. Unlike traditional centralized ML, our approach maintains data in a decentralized manner. Each hospital utilizes its own local dataset to train a model locally, which enhances privacy and security by ensuring that the raw medical data, which contains sensitive information, never leaves the local hospital environment.
- (iii) **Sensitive information:** FL aims to protect sensitive information by ensuring that this data never leaves the local entity (i.e., hospitals) to minimize the risk of data breaches or unauthorized access. In the context of healthcare data sharing, sensitive information includes personally identifiable information (PII) and PHI such as names, addresses, medical records, diagnoses, treatment histories, genetic data, and any other health-related details.

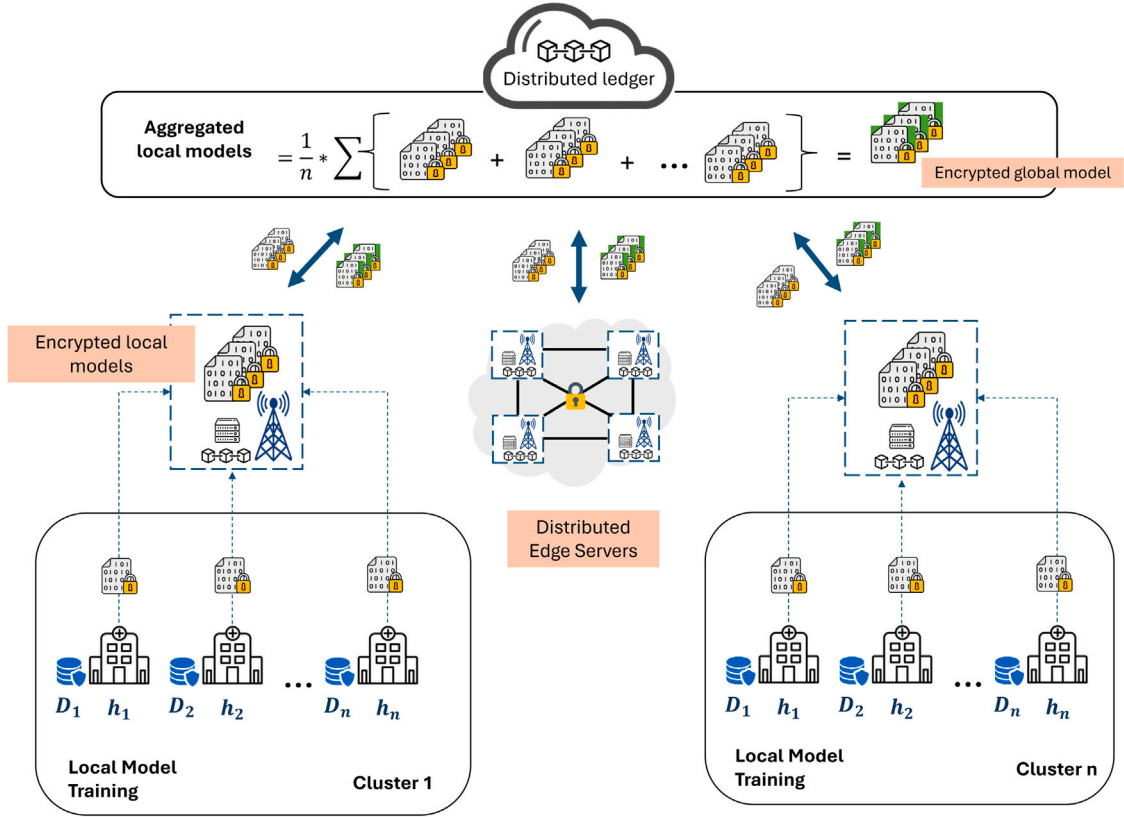


Fig. 2. Proposed model.

- (iv) *Trained local model*: Refers to a FL model that has been trained using data stored locally within hospitals. In this process, each hospital sends only the trained model's parameters to an edge server, where these updates are aggregated to create an enhanced global model.
- (v) *Clusters*: The hospitals are grouped into clusters to minimize communication overhead between the users and the edge server. Additionally, clustering participating hospitals based on geographic proximity, network conditions, or data distribution can reduce computational load and improve system performance. It is worth noting that each cluster is equipped with a blockchain-based edge server to collect the trained local models from the hospitals.
- (vi) *Edge servers*: The edge server is assigned to gather the trained local models submitted by hospitals and perform secure model aggregation. Additionally, edge servers facilitate local storage, communication, and processing capabilities and reduce communication and computation costs by enabling computational tasks to be performed closer to the hospitals, which act as data providers. In this setup, blockchain is utilized to establish distributed edge servers, ensuring the authenticity and integrity of transactions in healthcare data sharing, and facilitating decentralized reward mechanism.
- (vii) *Global model*: After training, each hospital sends only the updated parameters of its local model to the edge server within its cluster. The edge server then aggregates these updates to create the new global model. By leveraging homomorphic encryption, our approach allows the aggregation of encrypted local model updates without needing to decrypt them, thereby enhancing privacy and security in the system.
- (viii) *Trusted authority (TA)*: A trusted third party that responsible for initializing various security parameters, including generating and managing cryptographic keys. These keys are used to encrypt model updates exchanged between hospitals and the blockchain-based edge server. Additionally, the trusted authority authenticates the hospitals' identities to ensure that only legitimate and authorized hospitals can participate in the training process.

4.2. Design architecture and procedures

4.2.1. System initialization

Firstly, the initial set of model parameters (such as weights and biases for a neural network) is generated. The initial parameters of global model (ψ^{in}) are chosen based on the model architecture and the specific requirements of the application (e.g., a healthcare prediction model). Then, ψ^{in} are stored on distributed edge servers using blockchain smart contracts and shared with all participating

hospitals. Here, only authorized hospitals are permitted to join the system and download the initial model to participate in collaborative healthcare data sharing. The TA generates public and private key pairs (key_E^{pk}, key_E^{sk}) for blockchain-based edge servers to use in the data-sharing process. Similarly, hospitals are required to generate their own public and private key pairs (key_h^{pk}, key_h^{sk}) , which will be used to encrypt model updates. Furthermore, by employing a consortium blockchain, the edge servers are preselected and trusted, thereby enhancing the overall security and reliability of the system.

As illustrated in Fig. 2, we organize the set of users (i.e., hospitals) $H = \{h_1, h_2, h_3, \dots, h_n\}$ into clusters $C = \{C_1, C_2, C_3, \dots, C_n\}$, with each cluster equipped with blockchain-based edge servers $E = \{E_1, E_2, E_3, \dots, E_n\}$. In this context, the clustering mechanism is performed based on the similarity (δ) of the hospitals' data distribution using the cosine similarity technique [42], as follows:

$$\delta_{h_i, h_j} := \delta(\theta_i^*, \theta_j^*) := \frac{(\theta_i^*, \theta_j^*)}{\|\theta_i^*\| \|\theta_j^*\|} \quad (3)$$

$$C_i, C_j \leftarrow \arg \min_{C_i \cup C_j} (\max \delta_{C_i, C_j}) \quad (4)$$

where θ_i^*, θ_j^* represent the stationary solution [42] of hospital i and j . The system then separates the hospitals into groups such that the maximum similarity between hospitals from different clusters is minimized (see Eq. (4)). This approach effectively partitions the hospitals into groups that are more similar to each other.

4.2.2. Local model training and uploading

In every iteration t , each hospital h_i downloads the global model ψ^t and trains it locally using their dataset $D_i = \{D_1, D_2, D_3, \dots, D_n\}$. Here, D_i is not shared with other H , where $D_i \cup D_j = \phi, \forall i \neq j$, where $i, j \in H$. The training local models are computed as follows:

$$\psi_h^t = \psi^t - \eta_h \nabla F_h(\psi) \quad (5)$$

$$F_h(\psi) = \frac{1}{n_h} \sum_{x_i \in D_i} f_i(\psi) \quad (6)$$

where n_h is the total samples of h , f_i is a loss function for i th data point of h ($x_p^{(i)}, y_p^{(i)}$), $\nabla F_h(\psi)$ is the average of model parameters update, and η_h is learning rate. After training, hospitals obtain local model updates ψ_h^t and then encrypt them using HE and their public keys to ensure data privacy, as follows:

$$Enc_HE(\psi_h^t) = Enc(\psi_h^t, key_h^{pk}) \quad (7)$$

The encrypted local models $Enc_HE(\psi_h^t)$ are then uploaded to the cluster edge server E_i . Each hospital's model update is recorded as a transaction on the blockchain with hash function $hash\{Enc_HE(\psi_{h_1}^t), Enc_HE(\psi_{h_2}^t), \dots, Enc_HE(\psi_{h_n}^t)\}$, where transactions are validated by network nodes (i.e., distributed edge servers) $E_1, E_2, E_3, \dots, E_n$. It is worth noting that our system verifies performance by assessing the prediction accuracy of the trained local model [43] before proceeding with the model aggregation process. Therefore, only verified $Enc_HE(\psi_h^t)$ will be aggregated and stored for contribution assessment in the blockchain network.

$$E_i \leftarrow hash\{Enc_HE(\psi_{h_1}^t), Enc_HE(\psi_{h_2}^t), \dots, Enc_HE(\psi_{h_n}^t)\} \quad (8)$$

4.2.3. Model aggregation

After the local model training process, the blockchain-based edge servers E_i receive encrypted model updates $Enc_HE(\psi_h^t)$ from their assigned clusters C_i of participating hospitals and perform model aggregation. E_i provide local computation capabilities to perform the aggregation tasks based on homomorphic encryption technique. Then, E_i aggregate the model updates by averaging $Enc_HE(\psi_h^t)$ from all participating hospitals to generate the new global model through the following equation:

$$\psi_{gbl}^t = \sum_{h=1}^H \frac{n_h}{N} Enc_HE(\psi_h^t) \quad (9)$$

where N represents the overall number of data points (samples). Since the model updates are encrypted, the edge server does not have access to the raw data or the specific details of the model updates; it only aggregates the encrypted updates. By using homomorphic encryption as a secure aggregation technique, the edge server can aggregate model updates without decrypting them.

Furthermore, the new global model ψ_{gbl}^t is shared with all distributed edge servers E_i in the blockchain network. The distributed nature of the edge servers helps mitigate the risk of single points of failure, thereby enhancing the overall security of the FL framework. As a result, participating hospitals can download and decrypt the new global model, $dec(\psi_{gbl}^t)$, for the next round of local training at iteration $(t + 1)$.

Algorithm 1 A detailed of the proposed scheme. t is number of iteration; H number of hospitals h_n ; E number of edge servers E_n ; D^i is the local datasets; E_p is the number of local epochs; and η is learning rate.

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1: procedure EDGESERVER:
2:    $E_i$  initialize  $\psi^{in}$ 
3:    $E, H \leftarrow (key_E^{pk}, key_E^{sk}), (key_h^{pk}, key_h^{sk})$  ▷ Public-private keypairs generation
4:    $\delta_{h_i, h_j} := \delta(\theta_i^*, \theta_j^*) := \frac{(\theta_i^*, \theta_j^*)}{\|\theta_i^*\| \|\theta_j^*\|}$ 
5:    $C_i, C_j \leftarrow \arg \min_{C_i \cup C_j} (\max \delta_{C_i, C_j})$  ▷ Distribute the users into clusters
6:   for each round  $t = 1, 2, \dots, T$  do
7:     for each hospital  $i \in h_i$  in parallel do
8:        $Enc\_HE(\psi_h^t) \leftarrow \text{UserUpdate}(i, \psi^t)$ 
9:        $\psi_{gbl}^t = \sum_{h=1}^H \frac{n_h}{N} Enc\_HE(\psi_h^t)$  ▷ Global model aggregation with HE
10:    end for
11:  end for
12: end procedure
13: procedure USERUPDATE( $i, \psi^t$ )
14:    $Enc\_HE(\psi_h^t) \leftarrow \psi^t$  ▷ Executes on hospital  $i$ 
15:   for each local epoch  $j$  from 1 to  $E_p$  do
16:     for each batch  $b = \{x, y\}$  of  $D^i$  do
17:        $\psi_h^t = \psi^t - \eta_h \nabla F_h(\psi)$ ; ▷ Local model training
18:        $F_h(\psi) = \frac{1}{n_h} \sum_{x_i \in D_i} f_i(\psi)$ 
19:        $Enc\_HE(\psi_h^t) = Enc(\psi_h^t, key_h^{pk})$  ▷ Encrypt the trained local models
20:     end for
21:   end for
22:   return  $Enc\_HE(\psi_h^t)$  for aggregation
23: end procedure
24: procedure INCENTIVEUSER( $R_h^t$ )
25:    $E_i$  collects the list of participating hospitals
26:    $h_1, h_2, \dots, h_n$ 
27:   for  $h_1, h_2, \dots, h_n; E_i$  do
28:      $E_i \leftarrow \text{ConfirmTransaction}$ 
29:      $hash(\psi_{h_1}^t, \psi_{h_2}^t, \dots, \psi_{h_n}^t)$ 
30:      $Incentive \rightarrow R_h^t = contribution \cdot \psi_{gbl}^t$  ▷ Contribution calculation
31:      $R_h^t$  are distributed to  $h_1, h_2, \dots, h_n$ 
32:   end for
33: end procedure

```

4.2.4. Contribution calculation

Here, hospitals are incentivized through the use of blockchain-based smart contracts, which reward them for meeting the transaction requirements of the system model. The blockchain keeps track of the list of participating hospitals $H = \{h_1, h_2, h_3, \dots, h_n\}$ that contribute to generating the global model ψ_{gbl}^t , verifies their transactions stored in blockchain-based edge servers $hash\{Enc_HE(\psi_{h_1}^t), Enc_HE(\psi_{h_2}^t), \dots, Enc_HE(\psi_{h_n}^t)\}$, and calculates rewards based on their contribution to ψ_{gbl}^t according to the below equation:

$$Incentive \rightarrow R_h^t = contribution \cdot \psi_{gbl}^t \quad (10)$$

The iterations continue until the model either reaches a specified level of accuracy or exceeds a predetermined number of iterations. The detailed procedures of the proposed scheme can be found in Algorithm 1.

4.3. Security analysis

4.3.1. Eliminating centralized authority approach

Blockchain minimizes or replaces the role of a central server, removing a single point of failure and reducing vulnerability to attacks. In our framework, we utilize a consortium blockchain, where a group of pre-selected entities (i.e., edge servers) collaboratively manages the network, ensuring decentralized control while maintaining efficiency and accountability. Additionally, we employ a cross-silo FL setting, where decentralized data owners (i.e., hospitals) collaboratively train ML models without sharing raw data. Integrating blockchain into FL further decentralizes the process by securely recording model updates and ensuring consensus among participants, thus eliminating reliance on a central coordinator and reducing the risk of manipulation or tampering. This approach enables secure, privacy-preserving data collaboration across healthcare institutions.

Table 2
Simulation parameters settings.

Parameter	Specification
Processor	Intel® Core™ i7-7700 8 CPU @ 3.60 GHz
Memory	16 GB DDR4 RAM
ML library	Keras 2.15.0, TensorFlow 2.15.0, Scikit-learn 1.4.2
Extension library	Numpy 1.19.5, Pandas 1.3.3
HE library	Microsoft SEAL - Pyfhel 2.3.1
Programming language	Python 3.8.19
Blockchain environment	Ganache Truffle GUI 2.4.0
Dataset	COVID-19 radiography [44] and brain tumor MRI [45]
ML model	Convolutional Neural Network (CNN)

4.3.2. Secure computation and mitigation of inference attacks

HE allows computations on encrypted data without revealing the underlying plaintext, ensuring that patient data remains private throughout the model training and aggregation process. Furthermore, after local training, model updates or gradients are encrypted to minimize the risk of leaking sensitive information during collaborative learning, ensuring that only users with the corresponding secret key can decrypt the model. This preventive measure ensures that shared model updates cannot be reconstructed or exploited.

4.3.3. Reliable data security and integrity management

In the FL framework, data owners (i.e., hospitals) maintain full control over their sensitive data by storing it locally. Instead of sharing raw data, only model updates are uploaded to the blockchain network, ensuring that the data itself remains private and secure. To prevent dishonest hospitals from sharing invalid or corrupted data, all transactions are verified by the distributed edge nodes based on the prediction accuracy assessment. Thus, only the verified model updates are processed for global aggregation.

4.3.4. Auditable and transparent collaborative learning

Blockchain records all transactions (e.g., model updates and rewards) in an immutable distributed ledger. By using smart contracts, blockchain incentivizes healthcare institutions to actively participate in FL training. Rewards are distributed based on contributions to the global model, promoting fairness and collaboration. Furthermore, the decentralized incentive mechanism discourages dishonest behavior, as rewards are tied to verifiable contributions recorded on the blockchain.

5. Numerical results and discussion

5.1. Dataset preparation

In our experimental evaluation, we used public datasets for COVID-19 radiography and brain tumor Magnetic Resonance Imaging (MRI), referencing the work of [44,45], respectively. We selected these datasets to represent use cases that require collaborative handling, such as pandemic diseases and critical conditions needing immediate treatment. The COVID-19 radiography database was chosen for its relevance to the COVID-19 pandemic, where early and reliable detection is crucial for controlling the virus's spread and ensuring timely treatment. This dataset includes X-ray images of COVID-19 positive cases, normal lungs, and other lung infections like viral pneumonia and lung opacity (non-COVID lung infection). It has expanded to 3616 COVID-19 positive X-ray images, along with thousands of normal and lung infection cases. In our study, we focused on two categories, normal and COVID, specifically concentrating on the COVID-19 detection process to achieve our design objectives. On the other hand, the brain tumor MRI dataset used in this study consists of 7023 high-quality MRI scans of the human brain, categorized into four distinct classes: glioma, meningioma, pituitary tumor, and no tumor. This dataset provides comprehensive imaging data for both healthy and tumor-affected brains. In our work, we focused on two categories, no tumor and meningioma, aiming to develop and evaluate ML models for accurate tumor detection and classification, as well as to enhance treatment planning in the medical field. Fig. 3 illustrates the samples of dataset for COVID-19 X-ray image [44] and brain tumor MRI [45].

5.2. Evaluation setup

To implement our proposed scheme, we divided the process into two scenarios: HE-based FL for secure healthcare data sharing and blockchain-based FL to enable decentralized collaborative training. The Convolutional Neural Network (CNN) model was chosen for its effectiveness in image recognition, particularly in training models for COVID-19 and brain tumor MRI data, which are essential for medical imaging tasks. Fig. 4 provides an overview of the CNN architecture, which includes six convolutional layers, six MaxPooling layers, two fully connected layers, and a Softmax layer. A total of 352 kernel filters are used across the convolution layers. The first Conv2D layer has 32 filters of size 3×3 , followed by a MaxPooling layer with a pool size of 2×2 . The following layers use 32, 32, 64, 64, and 128 filters of size 3×3 , each followed by MaxPooling layers with a 2×2 pool size. Then, two fully connected layers with 128 and 64 hidden units lead to the final Softmax output layer, producing two outcomes: “normal” and “COVID” for COVID-19 X-ray images, as well as “no tumor” and “meningioma” for brain tumor MRI data.

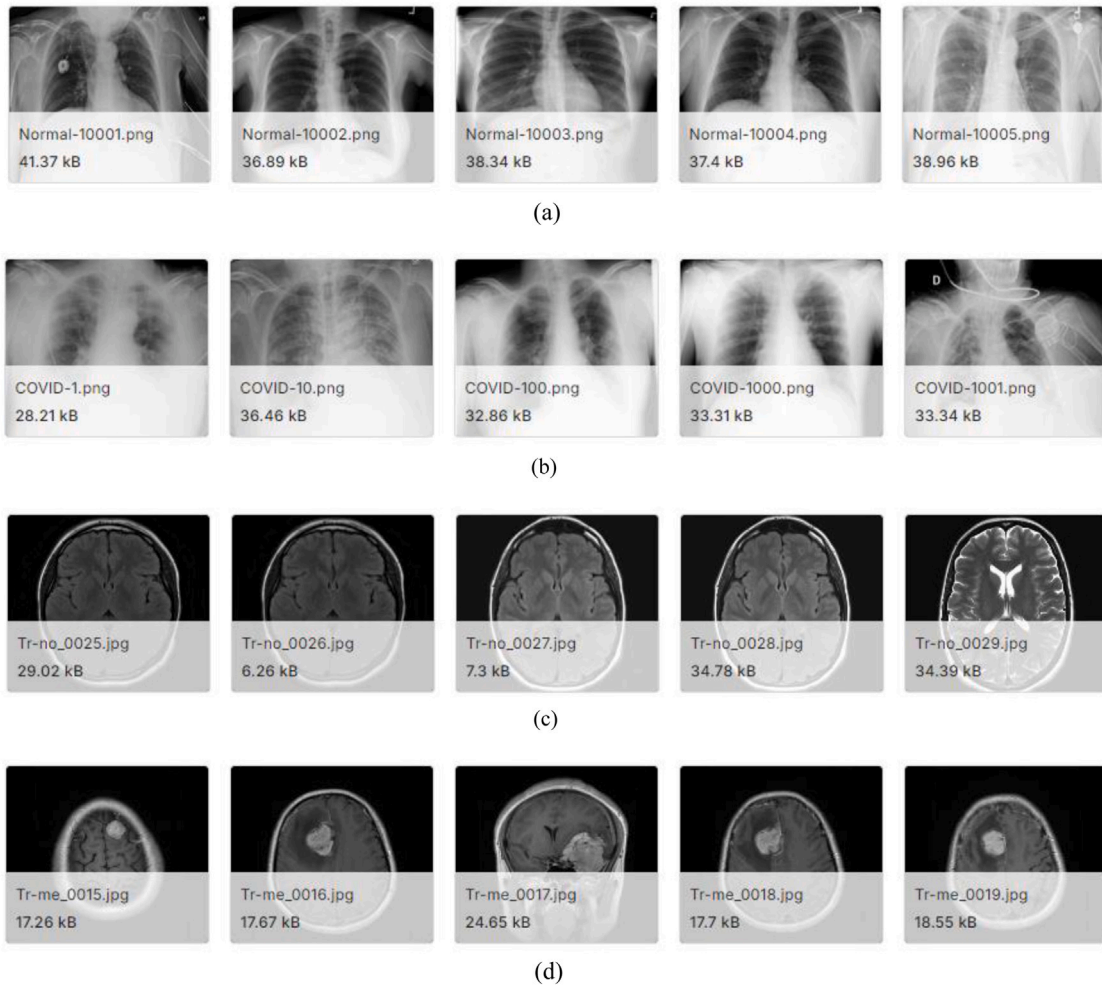


Fig. 3. The COVID-19 X-ray image [44] and brain tumor MRI [45] dataset samples: (a) COVID-19 negative X-ray image; (b) COVID-19 positive X-ray image; (c) Human brain MRI with no tumor; (d) Human brain MRI with meningioma.

The experiments were conducted using Python 3.8.19, leveraging the ML libraries Keras 2.15.0 and TensorFlow 2.15.0 for building and training the CNN models. Scikit-learn 1.4.2 was employed for additional ML functionalities, including model evaluation and metrics computation. Data manipulation and numerical operations were handled using Numpy 1.19.5 and Pandas 1.3.3, which facilitated efficient data preprocessing and analysis. For secure computation, particularly in the context of homomorphic encryption, we integrated Microsoft SEAL with Pyfhel 2.3.1, allowing for privacy-preserving model training and evaluation. Microsoft SEAL [46] is a homomorphic encryption library developed by Microsoft and released in 2015. Complementing this, Pyfhel [47] provides a user-friendly interface while maintaining access to the core functionalities of FHE. As a Python wrapper for Microsoft SEAL, Pyfhel features a layered architecture that facilitates access to homomorphic operations, key management, context parameters, ciphertexts, and serialization capabilities.

On the other hand, we deployed the Ganache Truffle GUI to create a local blockchain environment for deploying Ethereum smart contracts and transparently evaluating users' contributions. We used the default settings, operating on an RPC server in auto-mining mode, which included a gas price of 20,000,000,000 wei and a gas limit of 6,721,975 units. User addresses were generated through Ganache, with each address holding 100.00 ETH and being publicly accessible. The experiments and evaluations were conducted on a computer with an Intel® Core™ i7-7700 processor, featuring 8 CPUs at 3.60 GHz and 8 GB of RAM. The environment settings and specifications can be seen in Table 2, and the code is publicly available on GitHub.¹

¹ <https://github.com/blockchainflhe/Blockchain-based-Federated-Learning-with-Homomorphic-Encryption>.

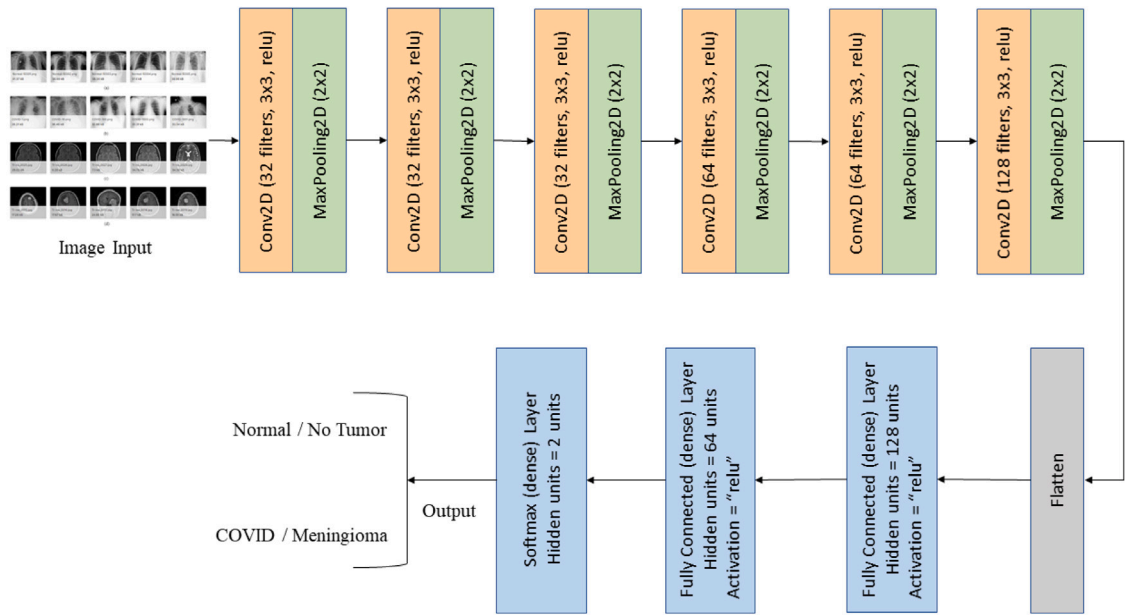


Fig. 4. The proposed CNN model architecture.

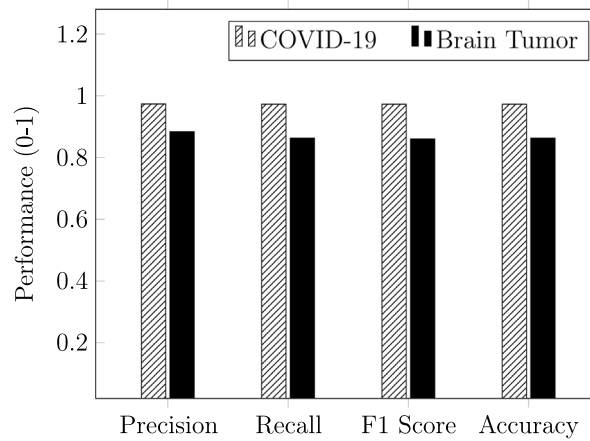


Fig. 5. The best performance of proposed model.

Table 3

The configuration of the training and testing datasets.

Configuration	COVID-19 datasets	Brain tumor datasets
$Scheme_1$	$D_{train} = 100, D_{test} = 30$	$D_{train} = 100, D_{test} = 30$
$Scheme_2$	$D_{train} = 400, D_{test} = 100$	$D_{train} = 400, D_{test} = 100$
$Scheme_3$	$D_{train} = 800, D_{test} = 200$	$D_{train} = 800, D_{test} = 200$

5.3. Experiment results

We started by selecting 1000 records from the original dataset for both COVID-19 and brain tumor MRI data, dividing them into 800 records for training and 200 for testing. We configured several parameters, including the number of epochs, batch size, and learning rate, set to 32, 10, and 0.001, respectively. Various metrics, including precision, recall, F1-score, and accuracy, were used to evaluate the system's performance. This simulation observed a procedure similar to that outlined in [16], aligning with our approach. Fig. 5 presents the best performance of our proposed model for detecting COVID-19 and brain tumors. For the COVID-19 dataset, the model achieved precision, recall, F1 score, and accuracy values of 0.9735, 0.9725, 0.9725, and 0.9725, respectively. Meanwhile, for the brain tumor dataset, the model achieved precision, recall, F1 score, and accuracy values of 0.8837, 0.8625,

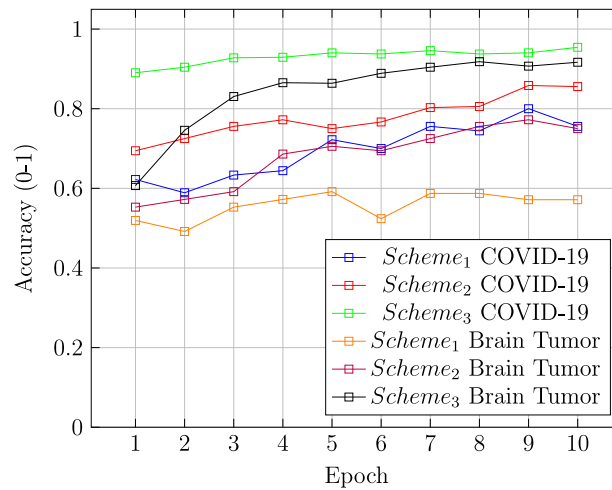


Fig. 6. The accuracy results from the varying data distributions across hospitals.

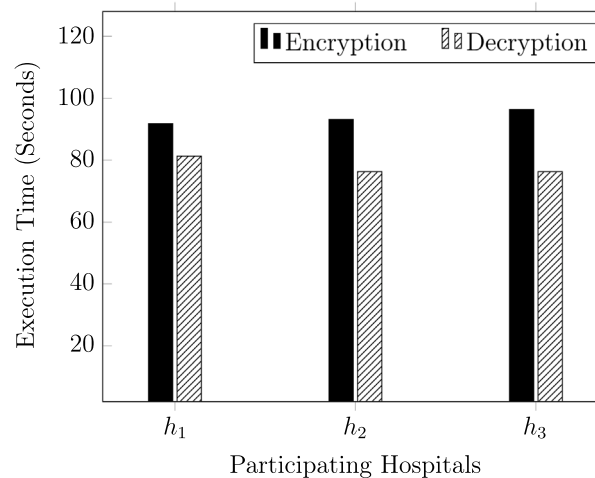


Fig. 7. The total time for model encryption and decryption.

0.8606, and 0.8625, respectively. We then varied the data distribution among hospitals to observe its impact on the accuracy of the local model updates. The configuration of the training and testing datasets is shown in Table 3. Fig. 6 illustrates the accuracy results obtained from varying data distributions across hospitals over 10 epochs. In this evaluation, *Scheme*₁, which used the brain tumor dataset, produced the lowest accuracy, while *Scheme*₃, utilizing the COVID-19 dataset, achieved the highest accuracy for the local model updates. In this sense, hospitals with larger datasets tend to achieve higher accuracy in local training. Conversely, hospitals with smaller datasets typically experience lower accuracy.

After local training, hospitals encrypt the local model updates using homomorphic encryption before uploading it to edge servers, and they decrypt the global model before the local training process begins. Fig. 7 shows the total time hospitals spend encrypting and decrypting the global model during collaborative training. As depicted in the figure, the decryption process is faster than encryption due to simpler operations involved, such as reversing encryption transformations or applying straightforward algorithms to retrieve the original plaintext [48]. On the other hand, Fig. 8 shows the overall execution times across clusters with different levels of data distribution. We configured five clusters: *C*₁ and *C*₂ exhibit almost homogeneous data distribution, whereas *C*₄ and *C*₅ show more heterogeneous data distribution, with *C*₃ positioned in the middle. As predicted, the execution time in *C*₅ is higher than in other clusters, especially *C*₁. In clusters with more heterogeneous data, such as *C*₄ and *C*₅, the variability between data points increases, causing the local models to diverge more from each other. This divergence leads to poor convergence and longer processing times during model training and aggregation. Consequently, the execution time in *C*₅, with the most heterogeneous data, is higher than in other clusters, particularly compared to *C*₁, which has more homogeneous data. Homogeneous data clusters converge more quickly, leading to shorter execution times.

We continued our observation by varying the number of hospitals in each cluster: *C*₁ with 2 hospitals, *C*₂ with 3 hospitals, *C*₃ with 4 hospitals, and *C*₄ with 5 hospitals. We performed the evaluation on both the COVID-19 and brain tumor datasets

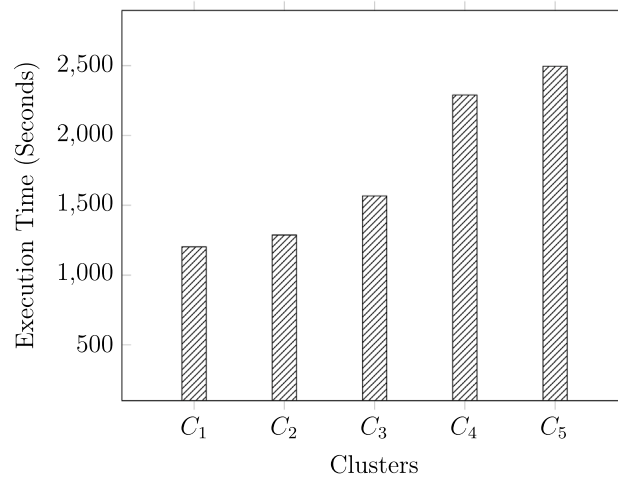


Fig. 8. Execution time in seconds with the various data distribution level.

Table 4

The results of system performance.

Cluster	Dataset	Precision	Recall	F1 Score	Accuracy
$C_1 \rightarrow H = 2$	COVID-19	0.86511	0.83750	0.83436	0.83750
$C_2 \rightarrow H = 3$	COVID-19	0.87212	0.84500	0.84212	0.84500
$C_3 \rightarrow H = 4$	COVID-19	0.90724	0.90000	0.89955	0.90000
$C_4 \rightarrow H = 5$	COVID-19	0.93859	0.93000	0.92965	0.93000
$C_1 \rightarrow H = 2$	Brain tumor	0.82608	0.73333	0.71292	0.73333
$C_2 \rightarrow H = 3$	Brain tumor	0.81127	0.76750	0.75902	0.76750
$C_3 \rightarrow H = 4$	Brain tumor	0.87537	0.85000	0.84742	0.85000
$C_4 \rightarrow H = 5$	Brain tumor	0.88368	0.86250	0.86057	0.86250

Table 5

The results of system computation time.

Cluster (COVID 19)	Total encryption time (s)	Aggregation time (s)	Total execution time (s)
$C_1 \rightarrow H = 2$	599.83940	119.42638	1330.64357
$C_2 \rightarrow H = 3$	899.26339	186.61607	1629.75875
$C_3 \rightarrow H = 4$	1160.82893	230.19174	1960.49950
$C_4 \rightarrow H = 5$	1655.71437	269.49947	2511.15039

Table 6

The effect of q and m to computation time.

Parameters	Total encryption time (s)	Aggregation time (s)	Total execution time (s)
$q = 128, m = 1024$	1655.71437	269.49947	2122.81493
$q = 192, m = 1024$	1660.53336	277.79007	2577.25304
$q = 128, m = 2048$	2282.26301	895.55570	4025.69411
$q = 192, m = 2048$	3079.10102	1207.83221	5929.70427

using homogeneous data distributions. Table 4 summarizes the results of system performance, where C_5 with COVID-19 dataset outperforms other clusters in every performance parameter: precision of 0.93859, recall of 0.93000, F1 score of 0.92965, and accuracy of 0.93000. In contrast, C_1 with brain tumor dataset achieves lower performance metrics: precision of 0.82608, recall of 0.73333, F1 score of 0.71292, and accuracy of 0.73333. Moreover, Table 5 depicts the results of system computation time among clusters. In this setting, we use the COVID-19 dataset with *Scheme₃* in Table 3. The results indicate that an increase in the number of participating hospitals leads to a proportional increase in system computation time. Additionally, we also adjusted two hyperparameters in the homomorphic encryption environment: the ciphertext modulus (q) and the polynomial modulus (m). q controls how much noise can accumulate before decryption fails [47], while m is associated with the complexity of the encryption scheme [46]. In our simulation, the value of q is set to 128 and 192, while the value of m is set to 1024 and 2048. Table 6 presents the effect of q and m to computation time. As a result, when $q = 128$ and $m = 1024$, the overall computation time is lower compared to other settings. In contrast, when $q = 192$ and $m = 2048$, the computation time is the highest. This indicates that larger values of q and m lead to increased ciphertext sizes and longer computation times.

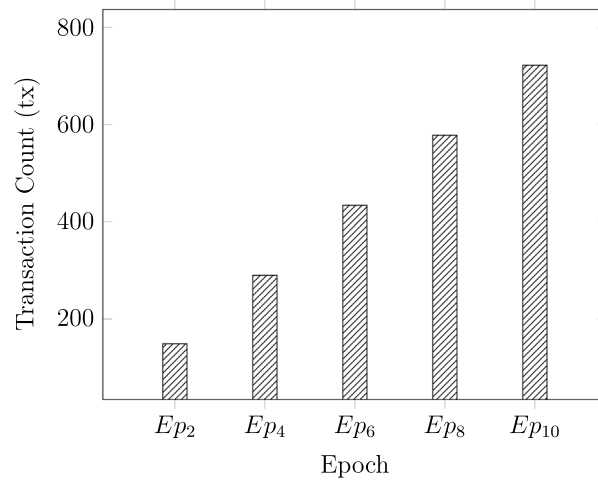


Fig. 9. The total number of transactions generated in the blockchain network.

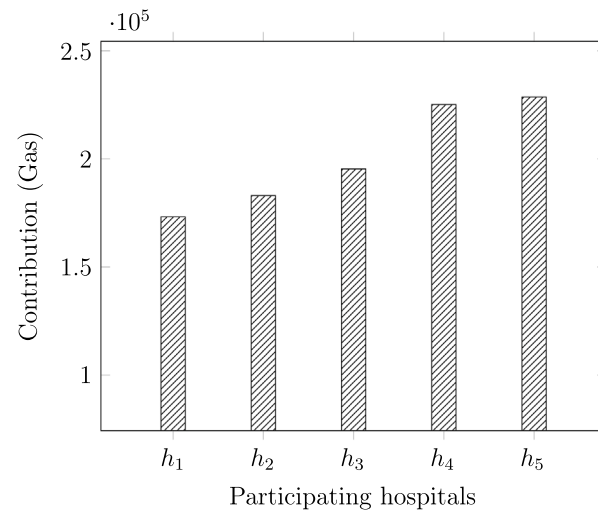


Fig. 10. The users' contribution results.

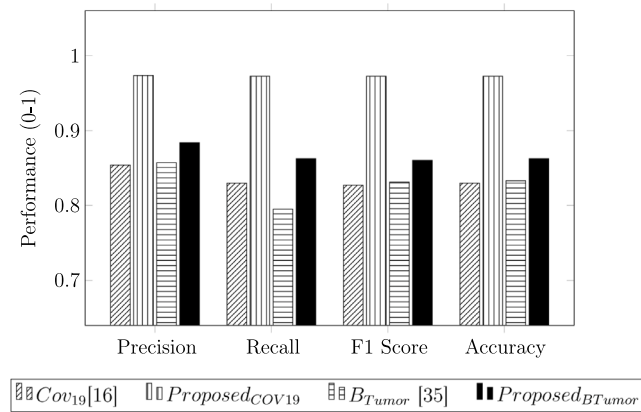


Fig. 11. Performance comparison with the existing works.

Once hospitals upload their local model updates to the cluster edge servers, these updates are recorded as transactions on the blockchain network. Fig. 9 illustrates the total number of transactions generated in the blockchain network across various epochs (Ep). Ep_2 recorded 149 transactions, Ep_4 generated 290, Ep_6 reached 434, and Ep_8 and Ep_{10} generated 578 and 722 transactions, respectively. On the other hand, Fig. 10 demonstrates how users' contributions are distributed to build the FL global model using the Ethereum platform. Five collaborative entities (hospitals) have been established to develop a robust framework for sharing healthcare data. This process entails gathering and integrating models from different hospitals within the proposed framework and storing them on a decentralized ledger blockchain. Following this, hospitals receive rewards based on their documented contributions to the blockchain's distributed ledger when a new global model is generated.

Lastly, we compare our proposed model with recent studies [16,35]. As shown in Fig. 11, the results indicate that our model outperforms the approach presented in [16] for the COVID-19 dataset and demonstrates superior performance compared to the work suggested in [35] for the brain tumor dataset. Our framework outperforms both studies in all key performance metrics, including precision, recall, F1-score, and accuracy. For example, in the COVID-19 dataset, the accuracy achieved by [16] is 0.8300, while our proposed framework achieves an accuracy of 0.9725. Similarly, for the brain tumor dataset, [35] achieves an accuracy of 0.8331, while our framework performs better with an accuracy of 0.8625. In summary, while both [16,35] focus on securing the FL process using homomorphic encryption, our proposed framework goes a step further by integrating blockchain technology, which strengthens data security, decentralization, and incentivizes collaboration. In our framework, we enhance security and privacy through both blockchain and homomorphic encryption, which not only secures the model but also ensures transparency and trust among collaborating healthcare institutions. This dual-layered security approach in our framework allows for more robust protection of data and model integrity, supporting a more collaborative and secure environment for model training, ultimately leading to more accurate predictions and improved overall performance.

6. Conclusion

This paper presents a secure approach to Federated Learning (FL) that utilizes homomorphic encryption and blockchain to address security and privacy concerns within the healthcare system. Our approach adopts a cross-silo FL framework suitable for healthcare applications. Blockchain technology ensures integrity and transparency by recording all transactions on a distributed ledger. Additionally, blockchain facilitates a decentralized incentive system design to encourage participation in collaborative training across healthcare institutions. Homomorphic encryption safeguards sensitive medical information during collaborative FL model training and aggregation, enabling computations on encrypted data without decryption, thereby ensuring privacy and security. Numerical results demonstrate that our scheme aids healthcare privacy protection and achieves intended design objectives with decent accuracy. However, future research should focus on critical areas such as the impact of heterogeneous data distribution among hospitals, which challenges model personalization and convergence. Moreover, developing robust defense strategies against potential FL vulnerabilities and attacks is crucial to ensure the security and integrity of FL systems.

CRedit authorship contribution statement

Muhammad Firdaus: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Harashta Tatimma Larasati:** Writing – review & editing, Writing – original draft, Validation. **Kyung Hyune-Rhee:** Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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